

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: GRU
group_sec: 100
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 1024
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 50
hyperband_iterations: 3
group_msec: 100
samples_per_sec: 10
forecast_horizon_sec: 4
forecast_steps: 40
timesteps: 60

GRU Multi-Step Training Report

Report Generated: 2025-05-21 13:35:17

Training Start: 2025-05-21_12-25-05

Training End: 2025-05-21_13-10-13

Training Duration: 45.15 minutes

Predictions Start: 2025-05-21_13-10-13

Predictions End: 2025-05-21_13-11-57

Predictions Duration: 1.73 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 2.0262

MSE: 6.2257

RMSE: 2.4951

R2 Score: 0.6190

Multi-Step Forecast:

Forecast Steps: 40

MAE: 0.0367

MSE: 0.0020

RMSE: 0.0442

R2 Score: 0.7919

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 10,265 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

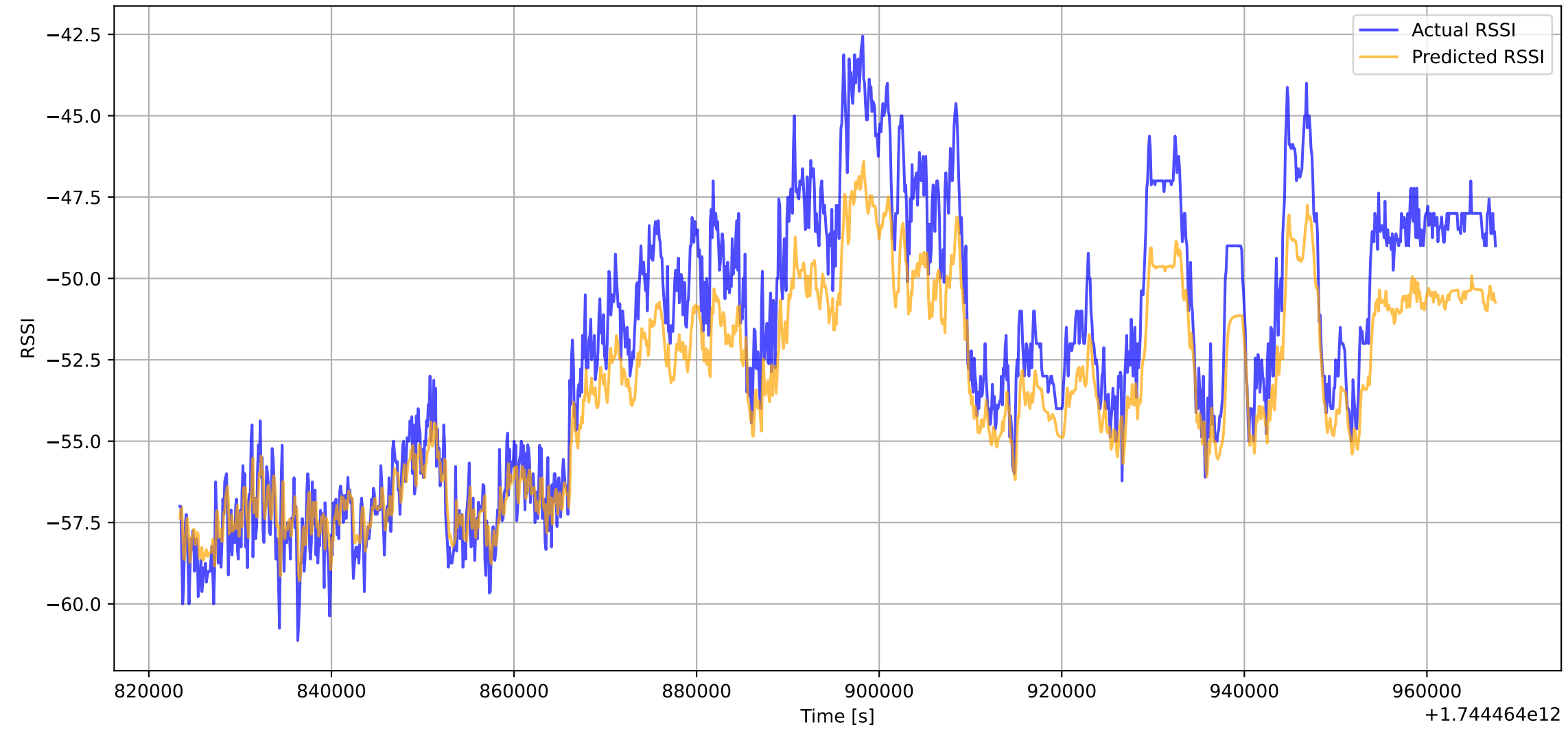
1744397085760
1744397085801
1744397086014
1744397086103
1744397086208

...

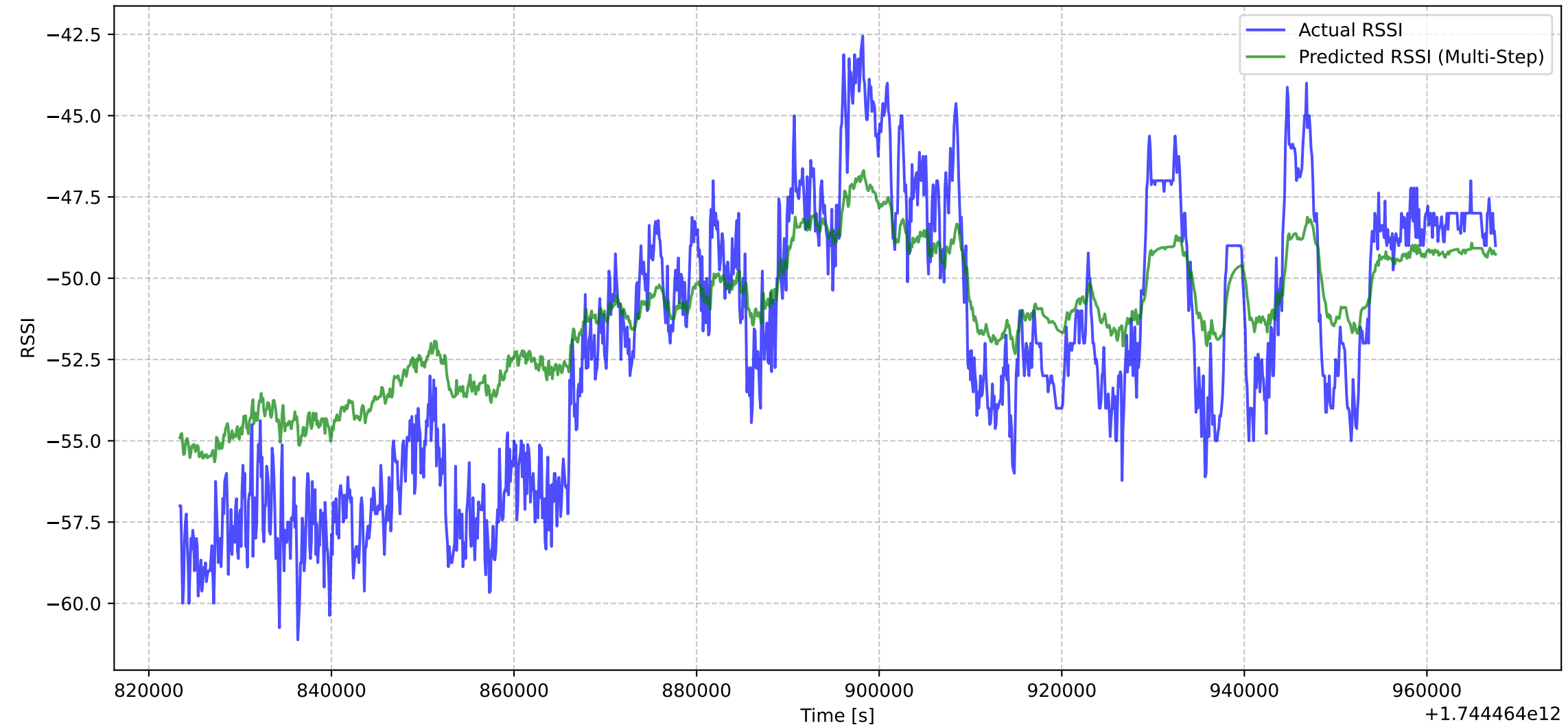
Last 5 timestamps AFTER:

1744464971004
1744464971101
1744464971209
1744464971306
1744464971402

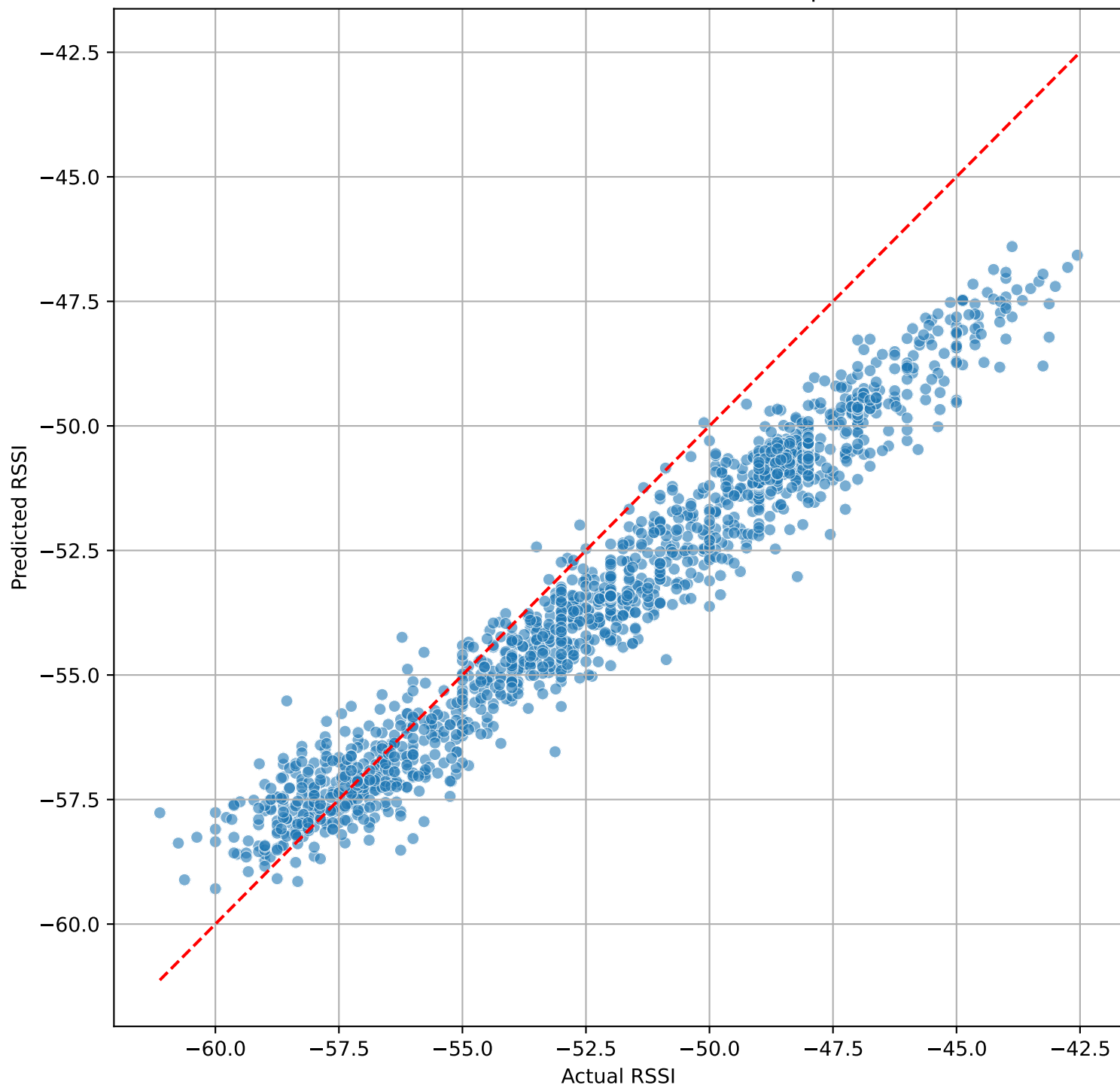
Actual vs Predicted RSSI (First Step)



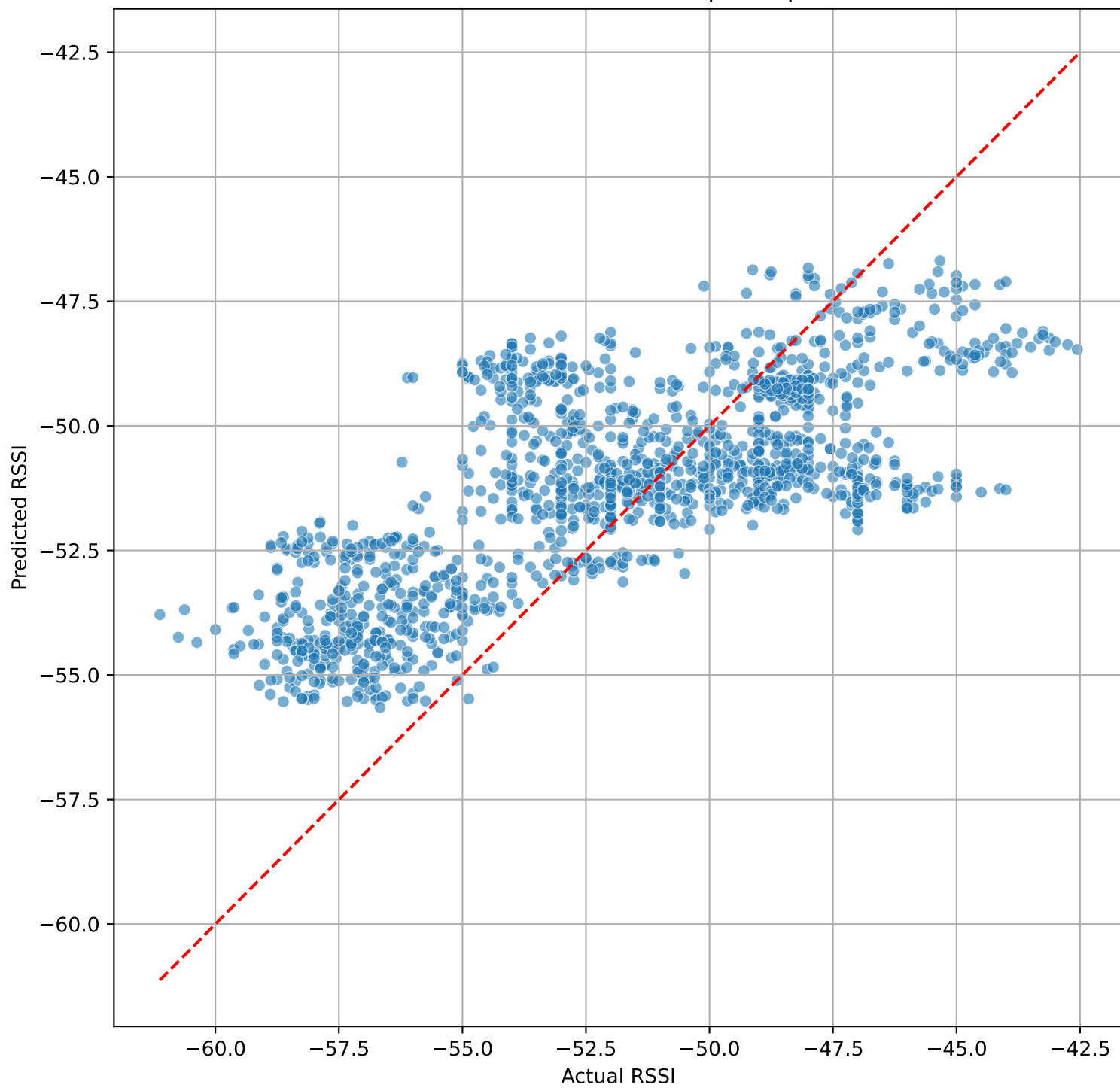
Actual vs Predicted RSSI (Multi-Step (step 40 $\approx$ 4.0 s))



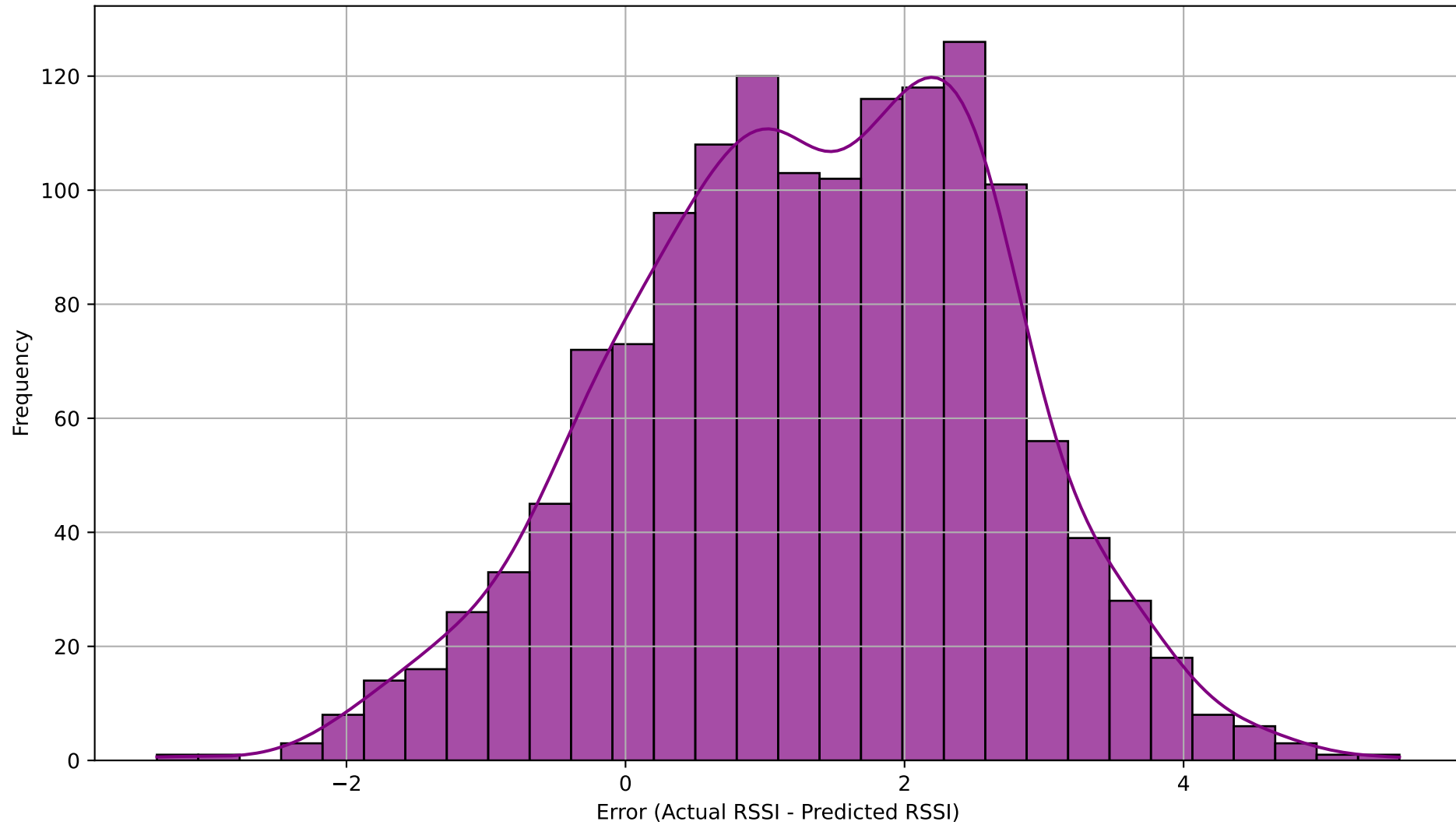
Actual vs Predicted RSSI (First Step)



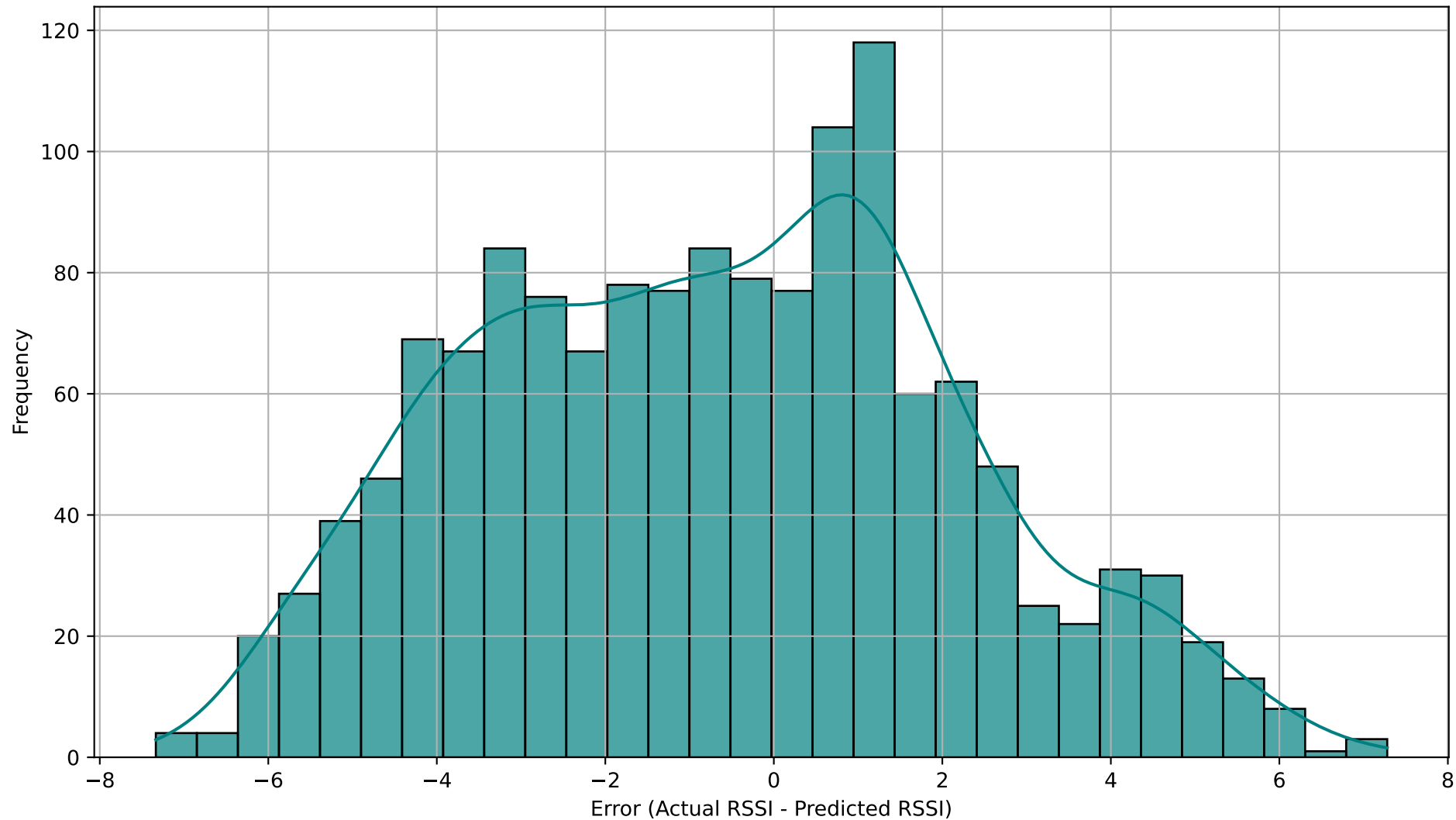
Actual vs Predicted RSSI (Last Step - step 40 $\approx$ 4.0 s)



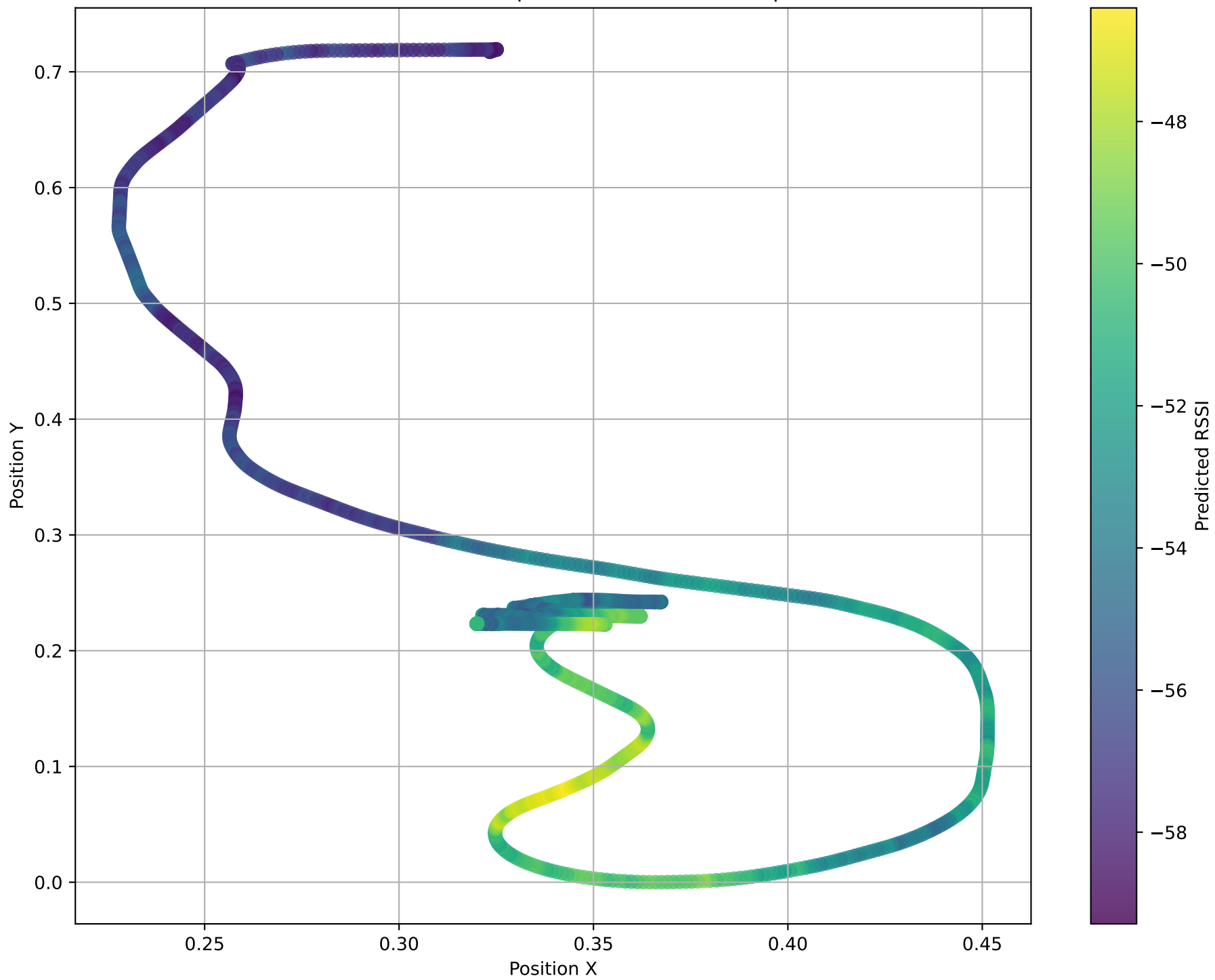
Distribution of Prediction Errors (First Step)



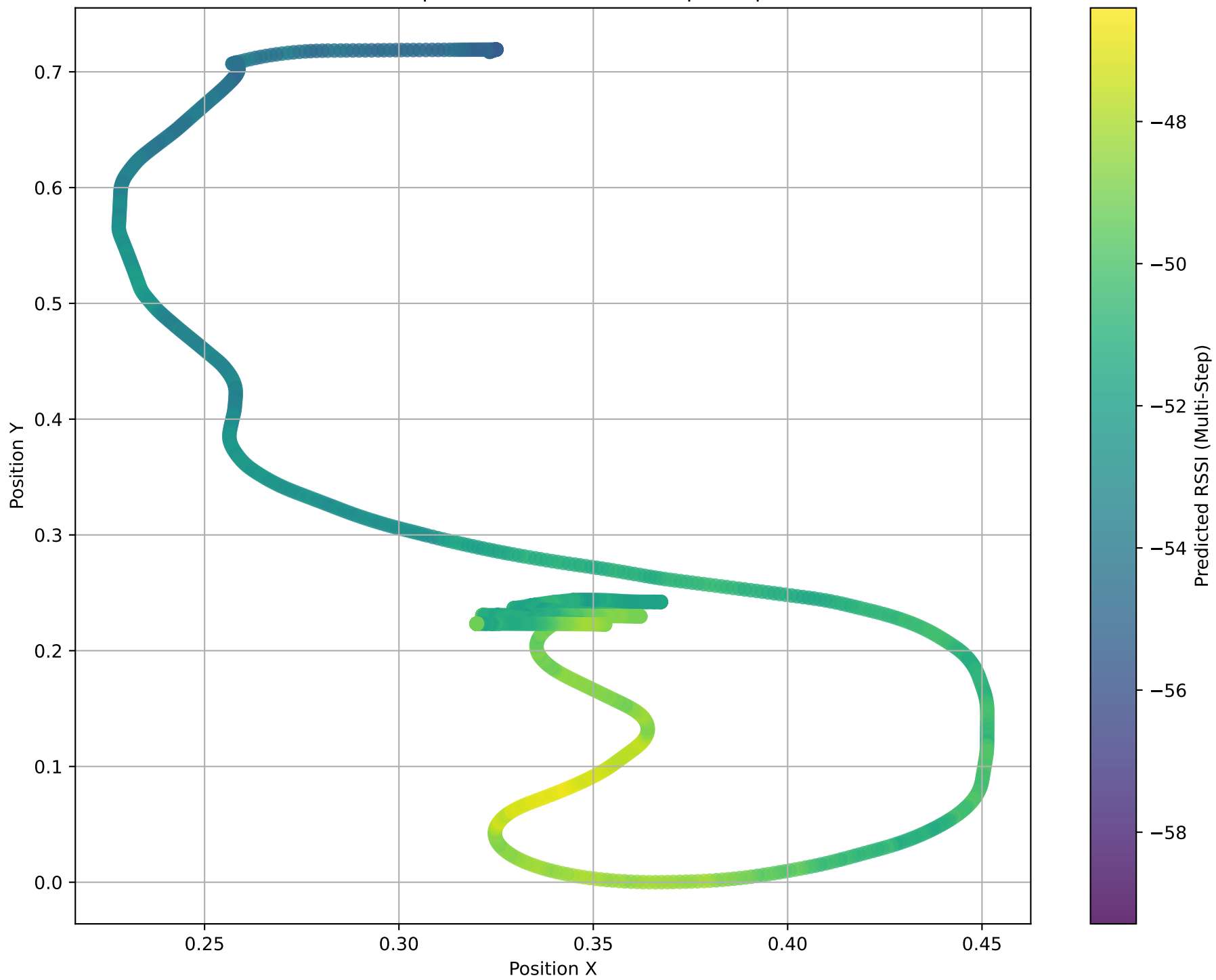
Distribution of Prediction Errors (Last Step - step 40 $\approx$ 4.0 s)



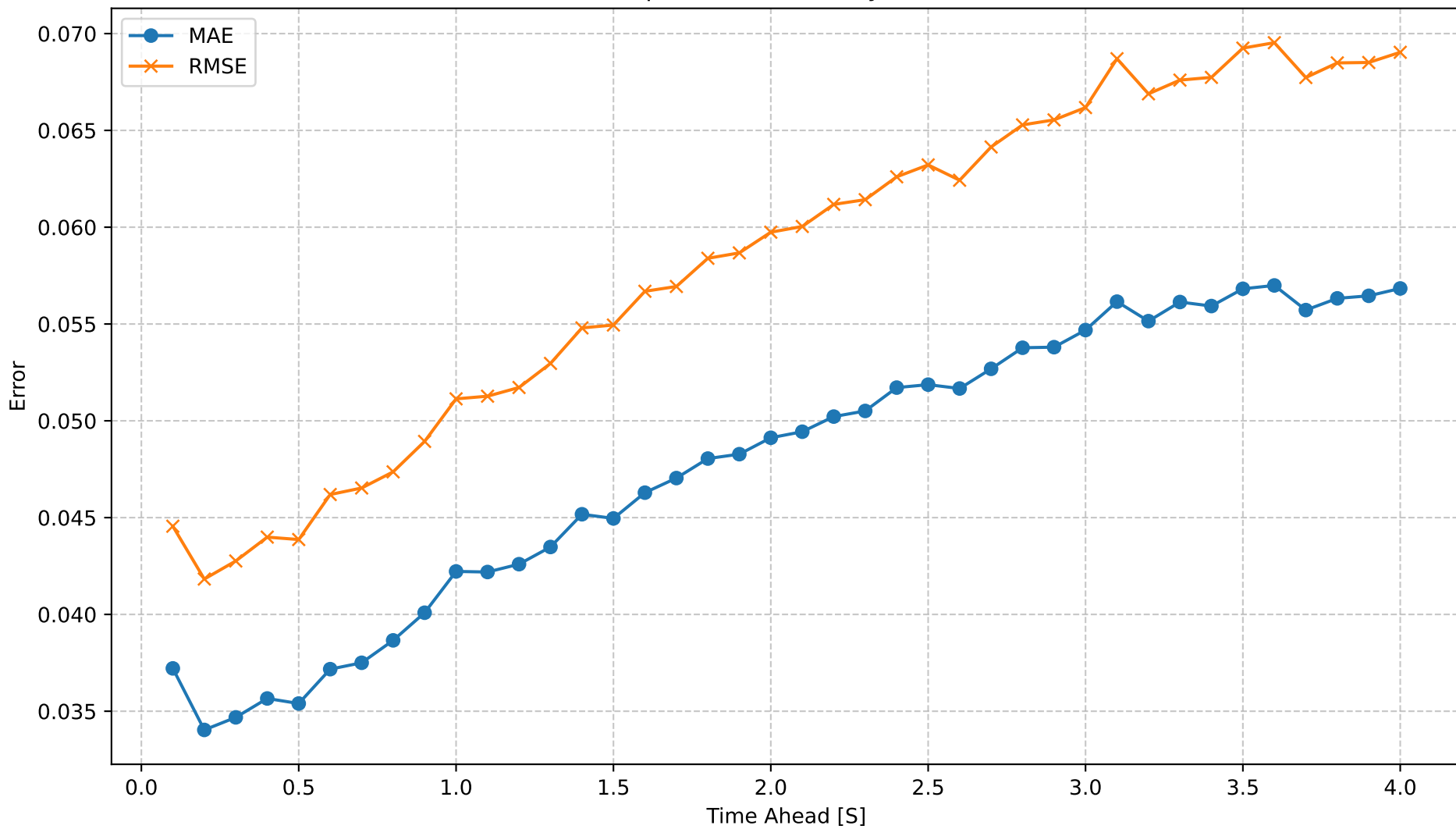
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: GRU

Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.04 / 0.04
- Mediana : 0.05 / 0.06
- 40º passo : 0.06 / 0.07

Crescimento do erro: 52.7% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 36 (0.06), 40 (0.06), 35 (0.06)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.